

Tomlinson, K., Jaffe, S., Wang, W., Counts, S. and Suri, S. (2025), ‘Working with AI: Measuring the occupational implications of generative AI’, *arXiv preprint arXiv:2507.07935* .

A1 Additional Tables

Table A1: Effect of LLM exposure on firm-level job listings

	Job postings
LLM exposure x Post	−0.010*** (0.002)
Firm fixed-effects	Yes
Industry-by-period fixed effects	Yes
Observations	1,268,013
Adjusted R ²	0.522

Notes: This table reports difference-in-differences estimates of the effect of LLM exposure on firm-level job posting activity. The dependent variable is an indicator equal to one if firm i posted at least one job vacancy in month t . LLM exposure is the firm’s standardized pre-treatment exposure score, calculated as the employment-weighted average of occupational LLM exposure scores from Eloundou et al. (2024) using the workforce composition on 30 October 2022. All regressions include firm fixed effects and industry-by-period fixed effects. All other definitions are as in Table 3.

A2 Heterogeneous Treatment Effects

A2.1 Firm-Level Employment Stocks

Table A2: Effect of LLM exposure on firm-level job outcomes, high vs. low compensation firms

	Employees	Low seniority	High seniority	Low seniority (%)	Av. comp.
Panel A: High compensation firms					
LLM exposure x Post	−0.007*** (0.003)	−0.008*** (0.003)	−0.005* (0.003)	−0.001* (0.0004)	195.240* (101.538)
Firm fixed effects	Yes	Yes	Yes	Yes	Yes
Industry-by-period FE	Yes	Yes	Yes	Yes	Yes
Observations	634,030	633,977	634,018	633,977	634,030
Adj. R-squared	0.975	0.973	0.974	0.963	0.956
Panel B: Low compensation firms					
LLM exposure x Post	0.0001 (0.003)	−0.001 (0.003)	−0.00002 (0.003)	−0.0004 (0.0003)	69.929 (46.938)
Firm fixed effects	Yes	Yes	Yes	Yes	Yes
Industry-by-period FE	Yes	Yes	Yes	Yes	Yes
Observations	633,983	633,978	632,313	633,978	633,983
Adj. R-squared	0.988	0.988	0.985	0.956	0.946

Notes: This table reports difference-in-differences estimates of the effect of LLM exposure on firm-level employment outcomes, separately for high- and low-compensation firms. Firms are split at the median pre-treatment average compensation level (October 2022). Panel A presents results for firms above the median compensation, Panel B for firms below the median. All other definitions are as in Table 3.

Table A3: Effect of LLM exposure on firm-level job outcomes, London vs. Non-London firms

	Employees	Low seniority	High seniority	Low seniority (%)	Av. comp.
Panel A: HQ in London					
LLM exposure x Post	-0.001 (0.004)	-0.002 (0.004)	-0.004 (0.004)	-0.001 (0.001)	214.807 (132.316)
Firm fixed effects	Yes	Yes	Yes	Yes	Yes
Industry-by-period FE	Yes	Yes	Yes	Yes	Yes
Observations	377,175	377,129	376,920	377,129	377,175
Adj. R-squared	0.982	0.981	0.981	0.978	0.975
Panel B: HQ outside London					
LLM exposure x Post	-0.004** (0.002)	-0.005** (0.002)	-0.002 (0.003)	-0.001** (0.0003)	71.962 (64.222)
Firm fixed effects	Yes	Yes	Yes	Yes	Yes
Industry-by-period FE	Yes	Yes	Yes	Yes	Yes
Observations	890,838	890,826	889,411	890,826	890,838
Adj. R-squared	0.983	0.983	0.981	0.979	0.979

Notes: This table reports difference-in-differences estimates of the effect of LLM exposure on firm-level employment outcomes, separately for firms headquartered in London, and those headquartered elsewhere. Panel A presents results for London-based firms, Panel B for firms based outside London. All other definitions are as in Table 3.

A2.2 Firm-Level Job Postings

Table A4: Effect of LLM exposure on firm-level job listings, heterogeneity

	High compensation firms	Low compensation firms	HQ in London	HQ outside London
LLM exposure x Post	-0.013*** (0.003)	-0.007*** (0.002)	-0.016*** (0.003)	-0.008*** (0.002)
Firm fixed-effects	Yes	Yes	Yes	Yes
Industry-by-period fixed effects	Yes	Yes	Yes	Yes
Observations	577,489	690,524	377,175	890,838
Adjusted R ²	0.486	0.554	0.517	0.525

Notes: This table reports difference-in-differences estimates of the effect of LLM exposure on firm-level job posting activity across different firm characteristics. Columns 1-2 split firms by median pre-treatment average compensation level. Columns 3-4 split firms by headquarter location (London versus outside London). All other definitions are as in Table A1.

Table A6: Effect of LLM exposure on occupation-level job outcomes, London vs. non-London

	Job postings (log)	Salary	Salary (log)	Salary (SD)
Panel A: Job in London				
LLM exposure x Post	-0.034** (0.017)	-562.818*** (139.355)	-0.010*** (0.002)	74.609 (132.525)
Occupation fixed-effects	Yes	Yes	Yes	Yes
Year-by-month fixed effects	Yes	Yes	Yes	Yes
Observations	16,734	16,734	16,734	16,625
Adjusted R ²	0.954	0.922	0.926	0.605
Panel B: Job outside London				
LLM exposure x Post	-0.040** (0.018)	-475.947*** (141.444)	-0.011*** (0.003)	-119.267 (111.131)
Occupation fixed-effects	Yes	Yes	Yes	Yes
Year-by-month fixed effects	Yes	Yes	Yes	Yes
Observations	16,812	16,812	16,812	16,799
Adjusted R ²	0.961	0.938	0.945	0.683

Notes: This table reports difference-in-differences estimates of the effect of LLM exposure on occupation-level job outcomes, separately for London-based jobs and jobs based outside London. Panel A presents results for job openings in London, Panel B for job openings outside London. All other definitions are as in Table 4.

A2.3 Occupation-Level Job Listings

Table A5: Effect of LLM exposure on occupation-level job outcomes, high vs. low salary occupations

	Job postings (log)	Salary	Salary (log)	Salary (SD)
Panel A: High salary occupations				
LLM exposure x Post	-0.057*** (0.022)	-782.371*** (184.426)	-0.013*** (0.003)	-354.844*** (131.069)
Occupation fixed-effects	Yes	Yes	Yes	Yes
Year-by-month fixed effects	Yes	Yes	Yes	Yes
Observations	8,413	8,413	8,413	8,412
Adjusted R ²	0.964	0.903	0.901	0.726
Panel B: Low salary occupations				
LLM exposure x Post	-0.020 (0.031)	-112.614 (179.137)	-0.004 (0.004)	240.391 (170.804)
Occupation fixed-effects	Yes	Yes	Yes	Yes
Year-by-month fixed effects	Yes	Yes	Yes	Yes
Observations	8,402	8,402	8,402	8,392
Adjusted R ²	0.961	0.817	0.860	0.436

Notes: This table reports difference-in-differences estimates of the effect of LLM exposure on occupation-level job outcomes, separately for high- and low-salary occupations. Occupations are split at the median pre-treatment average posted salary (October 2022). Panel A presents results for occupations above the median salary, Panel B for occupations below the median. All other definitions are as in Table 4.